Medical Physics Department

Physiology Lab.

2nd Class

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Lab 3

Packed Cell Volume (PCV)



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Packed Cell Volume (PCV) or Hematocrit Value

The PCV is defined The percentage by volume of packed red blood cells in a given sample of blood after centrifugation.

The increase in PCV value means increase the number of red blood cells per unit volume of the suspension

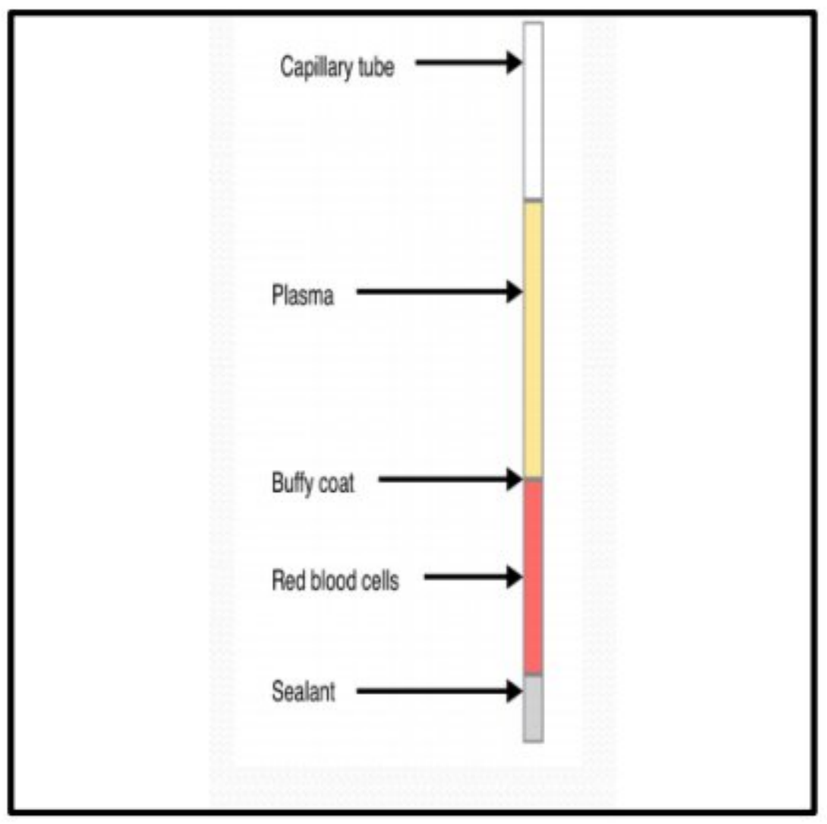
Uses

1- It used as a simple screening test for anemia.
2- To check accuracy of Hb value.
3- To calculate RBC indices such as MCV or MCHC.

Principle

When blood is collected in an anticoagulant and centrifuged, the blood will be separated into three

1. The mass of the erythrocytes at the bottom which is referred to as packed corpuscles volume (P.C.V).
2. White or gray layer of leukocytes and thrombocytes immediately above the red corpuscles mass that referred to as the buffy coat.
3. The blood plasma



Methods

There are two type of method use:

1. Macro-method using Wintrobe tubes.
2. Micro-method using capillary tubes.

The more popular one is the micro-method, as it has the advantage of **short time of centrifugation** and **better packing of the red cells**.

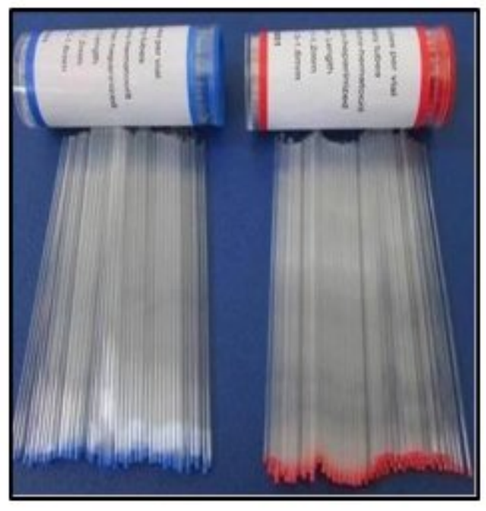
- **Material and instruments**
- Cotton
- Alcohol
- Lancet

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- Apparatus and reagent
  - Microhematocrit centrifuge.
  - Whole blood in heparin or EDTA tube
  - Microhaematocrit (capillary)
  - tubes
  - Plasticine clay
  - Microhaematocrit reader
  - Micro haematocrit reader.



Micro-centrifuge and reader



Capillary tubes



Clay



Specimen

1- Venous blood anticoagulated with EDTA or capillary blood collected directly into heparinized capillary tubes can be used.

2- Specimens should be centrifuged within 6 hours of collection

3- Hemolyzed samples cannot be used for testing.

Procedure:

1. Clean the area of the skin of a finger-tip or an ear lobe with a sterilized Alcohol cotton.
2. Prick the skin using the lancet.
3. Discard the first drop of blood, because it is mixed with tissue fluid.
4. Apply only gentle pressure beneath the pricked skin to help the flow of blood, because if more pronounce pressure is exerted, blood is likely to be diluted with interstitial fluid.
5. Allow the second drop of blood to be formed until it become large enough to fill 75% (3/4) of the heparinized capillary tube (heparin





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inside the tube) by the capillary action when it is brought closer to the blood.

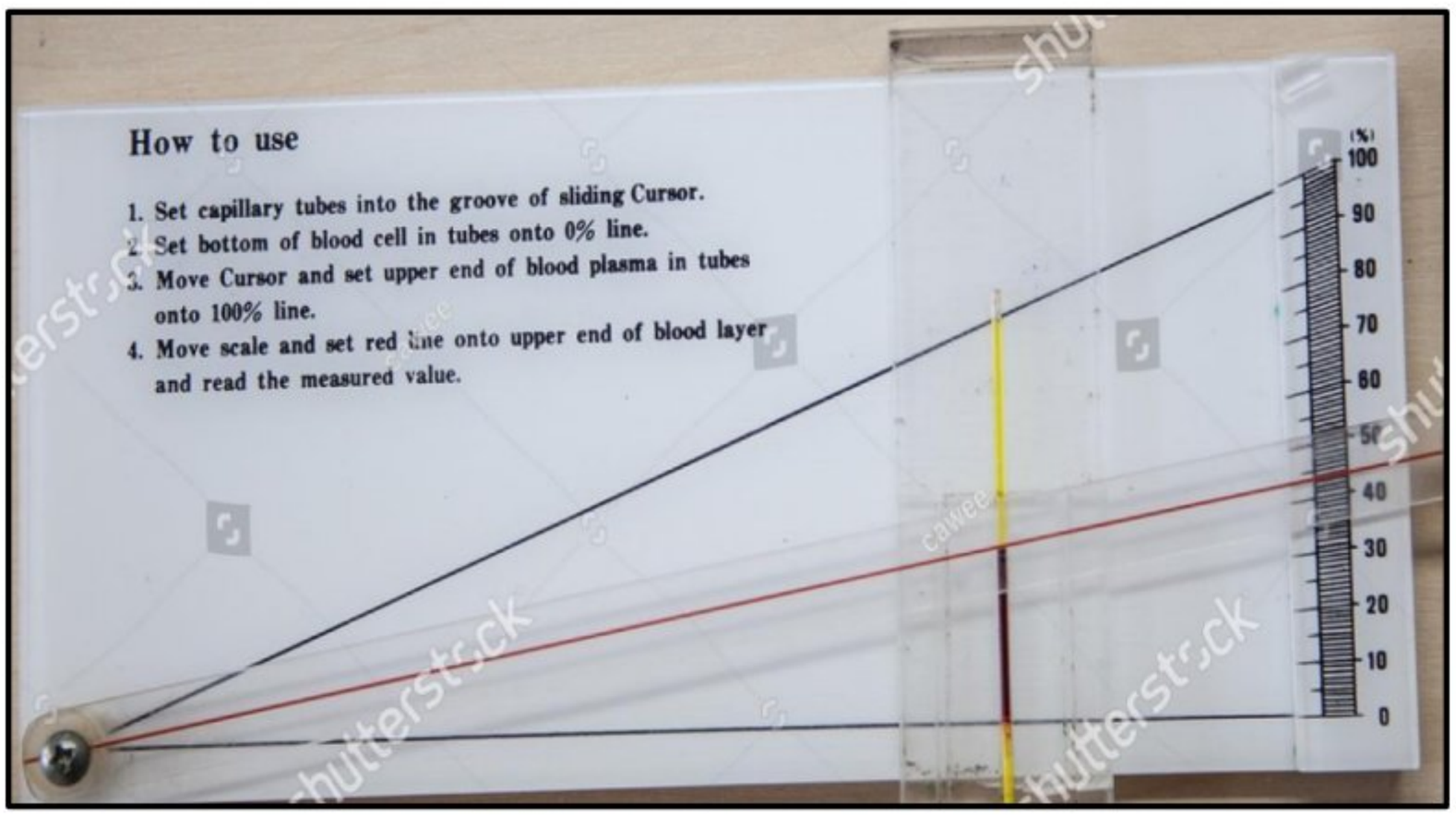
6- Wipe any excess blood from the outside of the tube.



7. Seal one end of the tube with clay or by fire. Avoid trapping air between the blood and plug.
8. Place the capillary tubes in the micro-hematocrit centrifuge radial slots or grooves. (The sealed end of the tube should point outwards), with balance the tubes in the centrifuge.



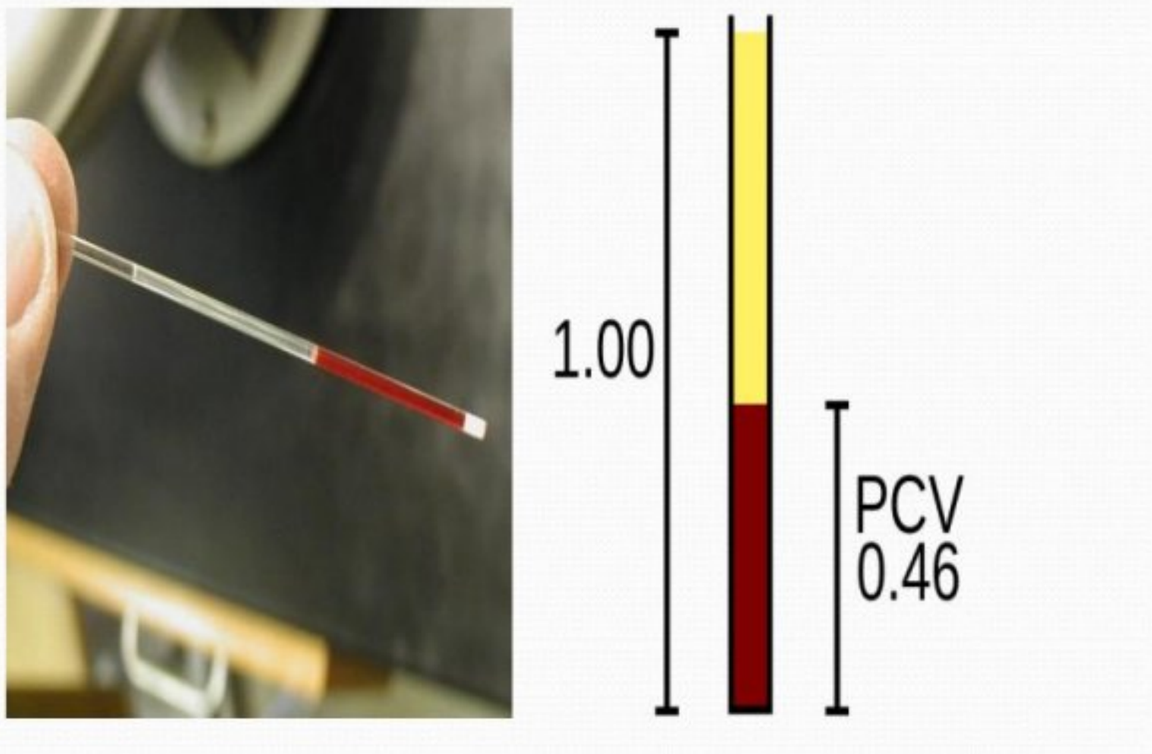
9. Close the cover of the centrifuge. Set the timer (most instruments require 3 to 5 minutes centrifugation time at 3000-4000 rpm). **Ideal speed approximately 10000 rpm for two minutes.**
10. The blood will have separated into layers. (Plasma / Buffy coat / Red cell layer).
11. Read the PCV (the RBC packing do not include the buffy coat) in the micro-hematocrit reader, the hematocrit result is expressed in percentage.



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How to read

1- Set capillary tube in to the groove of sliding cursor.
2- Set bottom of blood cell in tube on to 0% line.
3- Move cursor and set upper end of blood plasma in tube on to 100% line
4- Move scale and set red line on to upper end of blood layer and read the measured value.



Decreased PCV

1- bleeding
2- kidney disease
3- vitamin-B12 deficiency
4- Hemolysis
5- Bone marrow disorders such as leukaemia ,
6- lymphoma
7- Some medicines– including chemotherapy
8- pregnancy

Increased PCV

1-Severe dehydration
2-excessive use of diuretics
3-Erythrocytosis – excessive red blood cell production
4-Polycythemia vera – abnormal increase of blood cells
5-Hemachromatosis – an inherited iron metabolism
6-Disorder

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PCV Test Normal Range

The values of the PCV test normal range
are:

- Between 35% - 48%
- new born is 53%-65% while children is 30%- 43%
- females is 35.5 %- 44.9%, and
- males it is 38.3%- 48.6%

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